

“Where lots of people are sharing one thing, as soon as one person does something slightly different it can impact everyone” : A Formative Exploration of User Challenges and Expectations around Sharing of Accounts Online

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ABSTRACT

While users often share accounts with others, currently most accounts support only single user interactions. Within industry, Netflix and Disney+ providing profiles within accounts are testaments to popular services identifying and responding to user needs in this context but the solutions here are mostly naïve. Within academia, while sharing practices are of interest, the practicalities of dealing with them are yet to be studied. This paper highlights said gap in research and reports the preliminary findings of 4 user focus groups that reveal practical challenges and future expectations around the experience of sharing, social implications and user privacy, which when accounted for would support users in their sharing interactions. We intend to extend these findings by integrating them with expert interviews with ‘makers’ who research and work on said technologies, to produce a holistic set of design recommendations that form a practical guide for support around account sharing.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI).

KEYWORDS

Sharing, Accounts, Profiles, Data

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1 INTRODUCTION

The social ‘messiness’ of the real-world means that digital identities are not often neatly siloed and disparate, but they are, mirroring

offline social connections, often intersecting with each other. Considering the example of digital media platforms, it is not uncommon for users in a domestic setting to share their accounts [7, 10, 21] especially paid-for premium ones like Spotify Premium, with their friends, family or acquaintances. Also, prevalent here is the use of multiple accounts by a single user and shared profiles wherein a subset of the account is rendered to other parties who might need support (e.g., children’s profiles, carer-patient profiles etc.).

As Matthews et. al. [20] puts it, “we lack a systematic understanding of the nature of device and account sharing, across multiple device and account types”. Currently most technology platforms assume a single user to single account mapping though popular online platforms such as Netflix and Disney+ have started providing profiles within accounts, a sign of identification and response to user needs in this context but these solutions are still naïve and basic. They overlook the real-world complexities inherent in this scenario, thereby affording the users minimal support for their shared interactions, be it accounts intended as single user accounts or household accounts. What is apparent here is a stark gap in research, one that seeks the explication of the various socio-technical aspects of sharing. While research has, in siloed, disparate, and dispersed manners attempted investigation of various aspects of sharing practices [13] it is still limited in terms of its’ context, cohort and applicability. What calls to be explored are the practicalities of sharing, and for these practical reasonings to be representative of the real-world and thereby include users beyond the generic cohort of a single family or home to sharing between friends, family and acquaintances. Expanding research to co-located, remotely located shares, and users beyond adults, to include populations such as teenagers who are active yet dependant engagers of sharing.

This piece of work is a formative effort in responding to this gap as it reports the findings of a set of 4 user focus groups that explored these themes with adults who share accounts intended as single user accounts and household accounts within their families, outside the family in co-located scenarios and remotely located scenarios and a set of teenagers who share with their family and friends. The rest of this paper is fashioned as follows : Section 2 presents the research landscape around sharing practice, Section 3 details the research design of the focus groups and Section 4 reports a high level summary of the initial findings of the research. The findings uncover a set of user challenges around the experience, social implications and privacy and security concerns that users are currently navigating with much difficulty while also presenting a set of user wants and needs in near future sharing platforms.

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2 LITERATURE

2.1 Sharing of Technology

Sharing of digital artefacts range from passwords [17] to edge devices in the home [20] to calendars [19]. Here, several studies have reported on these sharing practices. Bryant and Campbell reported that 13.79% of students were most likely to be sharing their passwords with spouses or parents [6]. A wider study showed 67% of users in committed relationships to be sharing passwords to digital platforms with their partners [18]. While a number of these studies explored sharing of devices like mobile phones [3, 7, 16], others have looked into account sharing [1, 8, 12, 22]. These explored different application contexts and aspects of such sharing : Alghamdi et al. [1] and Singh et al. [22] looked into sharing of bank accounts, Egelman et al. [8] investigated computer OS-level profiles shared by families and Grinter et al. [12] explored similar aspects within Tivo profiles.

Thus, the practice of sharing digital resources online is a prevalent one among smaller social groups of colleagues, partners, families and friends [5, 8, 15, 20, 21] witnessing much popularity on digital media platforms and gaining increasing interest from media and HCI research [2, 11, 14].

2.2 Complexities of Sharing

2.2.1 Social. The social effects of sharing technologies have been of interest recently in research. Matthews et al. [20] through a multi-methods study of common account and device sharing practices lays out a taxonomy of six sharing types and two key themes: trust in sharing and convenience that cut across all sharing types. Jacobs et. al.'s work presents a detailed investigation of couples' sharing practices. They identified four sharing behaviours: intentional sharing, explicitly not sharing, unintentional access and unintentionally inhibiting access and classifies sharing information into three tiers : public content, tailored content and personal conversation. The work connects the content types to sharing practices to help support technologies in enabling sharing interactions better [13]. While their work helps add nuance in the explication of this complex domain, the practicalities associated with engaging in alternative design solutions that help support sharing is still a step away as they cite the potential of extending the work to include different social groups and special situations that might be representative of the real world.

Park et. al. reports a survey of 195 participants on Amazon Mechanical Turk who were asked about their relationship status and sharing behaviour. The work examines the temporal nature of account sharing practice at different stages of a relationship, for people of different ages and economic backgrounds. It presents a taxonomy around sharing behaviours and a set of design recommendations that help support users who are at three different stages of a relationship : "when they start sharing access with romantic partners; when they are maintaining that sharing; and when they decide to stop." [21] Another example of a move towards design guidelines is the work by Lampinen [18] which studied multi-person households that hosted couchsurfers. Couchsurfing accounts, while accessed by many is designed for a single user's engagement. The work found challenges around presentation of information to multiple persons and coordination challenges when

negotiating access to shared spaces. While this use case might be seen as an edge-case, it represents the diversity of the social aspects of sharing scenarios and the need to explore agendas beyond just the generic.

This work delves into these gaps left by studies that explored the social nuances of sharing accounts : of looking into the real-world practicalities of technologically enabling sharing practices while also extending the user group from just co-located families to co-located friends and housemates, remotely located family and friends and also teenagers.

2.2.2 Technological : Privacy and Security. Literature, along with social complexities of sharing, also uncovers the technological necessities and implications inherent in scenarios of sharing. Brush & Inkpen's study of 15 families showed that users, particularly in family situations, value personalisation over privacy in shared scenarios. The results here suggest a mixed model of the 'appliance' or 'profile' sharing type to help the technology be more aligned with the actual household sharing behaviours. Here, users ask for multiple passwords on profiles but again, this is more to do with relevance of content than privacy [5].

In their paper, Egelman et. al. [8] considered the notion of Family Accounts for shared home computers. The results showed Family Accounts to be an effective midway between a single account that is shared or having several individual profiles. The capability to switch between profiles seamlessly, without task interruption, showed that users switch to individual personal profiles only when they required privacy or personalisation.

When considering sharing of accounts, such as through sharing passwords, a key theme of interest in literature is the Security and Privacy (S&P) concerns. Watson et. al [23] focused on the dynamics of S&P in sharing scenarios by looking at how users navigate S&P decisions when sharing, how the sharing influences individual S&P attitudes and behaviours and the current mapping of S&P controls onto group needs. The work presents "why, how and what resources groups shared, their jointly construed threat models, and how these factors influenced group strategies for securing shared resources." [23]

These accounts currently tend to be dispersed, often focusing on different populations or use cases. Further, the generalisability of these findings in wider (or different) scopes and their applicability in other domains and use cases are yet to be considered. This research, works upon this gap to explore multiple populations that may engage with sharing of accounts while focusing on a popular domain for account sharing : media platforms, to help the findings be applicable to a certain practical context, yet be generalisable to wider scopes.

3 RESEARCH DESIGN

The research began with a set of 4 user focus groups. One group of families who share accounts within the same house, one group of users who share remotely, one group of users (not necessarily from the same family) who share accounts within the same house and a group of teenagers who share with family and friends. This being a formative experiment which aimed to study the landscape, the 4 groups were designed to include diverse kinds of sharing, of accounts intended as single user accounts and household accounts:

within and without family, remote and co-located and scenarios of support and dependency (teenagers). The diversity in opinion was explicitly respected here to ensure richness of data and response to a vast spectrum of users when the research progressed to the next design phases.

3.1 Participants

Twenty five participants took part in the sessions (numbers per session, 7: within family, 4: shared house but not family, 7: teenagers, 7: different houses). Of these thirteen were male, eleven female and 3 non-binary. Average age was 35 years (SD 10.5 years). Seven were 18 years or under, and the rest above 18. Most participants were residents of United Kingdom and one from Europe (Netherlands) and consisted of a diverse mix of professional backgrounds which included students (high school and University), researchers, movie director, sociologist, film producer, homemaker etc.. The participants were recruited through emails and advertisements marketed through the University's and the researchers' mailing lists, contacts and social media. A number of participants were also recruited through snowballing by family members who were recruited (particularly for the family group). The cohort consisted of a diverse group of individuals in terms of background. Again, this diversity of age, gender and social backgrounds were consciously chosen to make the findings representative of the wider real-world user population, helping highlight the nuance and sensitivities in sharing in as much detail as possible.

All participants were served ethics consent documents approved by the University Ethics Review Committee before they participated in the session. For teenagers, consent was derived from both the participants and their guardians. A remuneration of £30 was provided to all participants in exchange for their time and feedback.

3.2 Focus Groups

The sessions were facilitated by two researchers on Microsoft Teams. Each session included an introduction from the facilitators about the session aims followed by 3 sections: Sharing Accounts; Increasing Agency; and a Near Future scenario. Sharing Accounts covered what aspects of sharing participants found easy and difficult online, or if they ever felt like not sharing. The Increasing Agency section asked the users if they would like to share more, what they would like to be able to do with respect to sharing that they cannot do now, and what is stopping them from doing it. In the Near Future section users watched a video of a prototype Cross Media Profiler that enables users to import their profiles from various digital media platforms (e.g., Spotify, Netflix or Instagram) to help drive a recommender that would be significantly more powerful. The participants were asked to consider how their sharing practices would affect such a system, what features they would want to help them share, and any concerns around sharing while using the profiler.

The sessions lasted 90 minutes with each part of the agenda lasting around 25 minutes each along with a 15 minute Introductory session. Each session was led by 2 researchers who were guided by a script and slides that were pre-designed to support consistency in sessions. The sessions were audio recorded, transcribed and analysed qualitatively using thematic analysis supported by grounded theory [4, 10] to bring together the common underlying themes

that spread across these diverse groups. Participant responses are labelled with a number, and letter to delineate group: F for family group; S for shared house; D for different houses; and T for teenagers. The quotations presented here are not representations of the opinion of that particular group alone but a vignette into the conversations, that was chosen to best represent the respective wider theme being elaborated upon.

The results highlighted a number of user challenges and future expectations around technologies enabling support for sharing accounts and profiles. An outline of these challenges and expectations are presented in the following sections.

4 CHALLENGES

4.1 Experience

D3: I've actually gone so far as to kick my little sisters off of my Spotify because, as a little treat to myself, every month I pay for Spotify Premium, and I'm the only one in my family who does. So I said to my little sisters, "I'll share with you," and they immediately started adding so much bad music to my playlists (laughs). They were messing with my playlists, and that's the kind of problems that arise.

While the need for efficient personalisation and being served relevant content stands strong, also voiced was the importance of going beyond recommendations generated from history to those that are novel, serendipitous and that which allow for exploration, especially when sharing an experience or in a group.

Here, users attributed a number of challenges to company priorities, which they felt were often against users sharing these platforms. Users feel that this inherent interest of the makers to constrain sharing is resulting in a number of service-led experiential challenges that is making their current sharing practices difficult. The first of these are usability challenges around sharing accounts and profiles. These challenges include confusion around which bank cards are used by whom in which transactions, conversations and mechanics around changing a shared password or even the settings and preferences on a shared account, inefficient syncing of actions by different users and accounts e.g. inaccurate read flags on messages and emails on a shared account and conversational agents not recognising all members effectively, language barriers on subtitles, content availability restrictions across countries etc. But, most popular of these challenges is the limit of devices and users these accounts pose, a restriction which causes financial and social barriers for users. Also associated with the same constraint is the limit put on simultaneous experience consumption on a single account.

4.2 Social challenges

F4: “Where lots of people are sharing one thing, as soon as one person does something slightly different it can impact everyone.”

This could even lead to social tensions and need for negotiations which require further investment from the users' side.

Some of the highlighted social challenges here include the lack of hierarchy in the shares (or associated profiles). When all users

are given the same affordances and capabilities, irrespective of their role within the account, the interactions and resulting experience are confronted by challenges. Also identified here is the current lack of norms and clarity in terms of how an account could be shared, which is expected to guide users with their sharing practices. These hierarchies and guidelines to practicalise sharing through easier and smoother practices become keys to navigating the ephemeral and complex social scopes the users have to manage. Socially derived challenges here include the inherent temporal nature of relationships [21] and the resulting share practices, and limitations set on the type of platforms used due to the preferences of others involved [13]. Our results show that especially when elders are involved, users reported to often having to continue using traditional platforms that the others are more comfortable with when they themselves would have preferred a move to a more novel alternative.

And finally, strongly associated with social tensions were the financial dependencies of these shares which are often done for convenience [20]. This is especially for teenagers who often shared for financial reasons and did not want to share any more than they already did. Here, users highlighted the challenges of sharing financial resources on online platforms, especially credit cards and bank details, when shares require more mechanisms for control and management within the profiles to be able to mediate and approve transactions in frictionless and transparent manners.

4.3 Privacy and Security Challenges

F1: For example, we were remodelling the bathroom and I kept it to myself, and I made it a private [Pinterest] board. But she sees all my boards, even the private ones. So she quickly was criticising all the pins I did because she has access to them. So I did lose that privacy because I shared this account.

Such possibility of lack of privacy in shared accounts makes several social tasks like deciding on screen time allowance for children, buying gifts and planning surprises and even 'closeted' lifestyle choices like sexual orientation more challenging endeavours.

The concern of security is heightened when an account is shared across people and locations. Here, highlighted were two aspects of security. One is around the sharing of passwords, a common practice when an account is shared [6, 16, 18]. Users identified the extra vulnerabilities they face when sharing a password, especially if they use the same or similar passwords on other accounts. Users resort to using temporary and/ or 'simpler' passwords for ease of sharing which in turn makes the entire account more vulnerable to threat. Also, the notion of security was expressed to be perceived differently by different people and hence sharing an account, especially with those that do not value security measures compromises account security for everyone, a situation also highlighted through the need for group stewardship in S&P by Watson et. al[23].

The other aspect of security is around the management of security itself. To continue the example of passwords, when sharing an account there would be several occasions where people might be added or removed and subsequent changes to passwords and other data on the platform. Here, currently there are no efficient methods of technologically mediating these scenarios of change.

5 FUTURE EXPECTATIONS

5.1 Experience enhancement

A key aspect of experience enhancement is associated with the personalisation of the experience itself, one that was valued beyond privacy in previous research [5]. When sharing a profile with multiple users, using the shared data to resolve the context of the present session and deliver appropriate recommendations that are relevant to the users become of paramount interest.

F6: And then also thinking about the little thing I said at the end there about the context missing, whether or not but again, this gets into this having to configure all the time. It would be quite interesting as to, you know, I'm at work on my own and that's your profile. Or I'm watching TV with [name], or we're with the kids, or whatever, and it kind of understands the context of who you're with and it can put all of that together to then kind of create and deliver those recommendations?

Also prevalent here are multiple devices that share the same account. Many of these devices might be used for specific purposes or monopolised by certain individuals. Firstly, affirming previous research, the need to easily switch between these devices becomes a key user priority [9]. Secondly, to be able to segregate such usage, however minimal or extensive and not mix up different users' data and interests by enabling device level control and other context resolution strategies on sharing of data was cited as useful here. Synchronisation of the experience and updates on these several different devices and accounts also becomes a priority wherein changes made by or with respect to one user is reflected in effective manners to other users and on other devices effectively.

A significant step towards alleviating the concerns around the aforementioned complexities, were aggregators : platforms that help bring together different profiles and accounts. But, irrespective of whether an aggregator is provided or not, the users request for better standardisation between platforms so that users are aware of how to navigate their sharing practices on their various different accounts and devices.

5.2 Social requirements

The immediate social implications of the experience began with expectations around sharing the experience. Sharing of the experience here could be the ability to make recommendations to others on the account, through the account, particularly within media platforms, it could be users coming together to curate lists and recommendations for their shared sessions or being able to look into others' watch history and activity and make decisions for themselves. Also discussed in tandem was the possibility of the system bringing together data from profiles to generate shared recommendation lists.

D6: It would be nice to have some kind of list between the Netflix profiles and a common list, then we can add our favourite movies or what's next to watch between the profiles because I always talk with my boyfriend, "We should watch this later," but we always forget which movie it was. It would be nice to make a

list between the profiles of what to watch later, then it would be quite easy to pick up one of them and watch it together.

Another key area of immediate experience sharing was around sharing of a session. Here, more online support for the easy enabling of shared sessions like watch parties was highlighted. Examples of such measures included everyone having a shared code for the session and inputting the code, which would temporarily pull together the associated profiles for shared recommendations.

While the above might be easier when all users are in the same geographic location, remote sharing of experience is currently a much bigger challenge. Presently, users are forced to find informal ‘hacks’ that are less than efficient and maybe even against the terms and conditions of these platforms. Online affordances that make shared consumption of experiences and usage of accounts more seamless remotely, even on an international scope also becomes a priority here.

5.3 Privacy

T1 : Sometimes I think I’d kind of like to know what I’m sharing. I know like sometimes companies like really try and hide what they’re sharing about you and with what T6 says about just being able to set up profiles in accounts and then being able to manage that and actually know what everyone is sharing and like what data is actually being controlled and not just everything is out there mysteriously.

Agency transcends beyond transparency to be able to make changes to the data and the resulting experience. This is not just about the experience, but when the account is shared it almost becomes a social need for privacy and not being judged by others for the kind of interactions one might be engaging with. These proposed agency measures ranged from being able to delete items from history to deciding who to share what with on the platform to T6: “being able to select what bits I share and what bits I don’t want to share at certain points.” to being able to delete entire profiles, if required.

When considering shared online spaces as proposed by Lampinen [18], pronounced within this scope was the need to maintain privacy within them. This would be enabled through passwords and codes on the individual profiles or shared sessions and allowing for private spaces for each party, where others would not have access.

One key technological solution to allowing sharing but in a private manner which also embodies easy and useable interactions for all the users is allowing for multiple passwords for the different users involved. That way users are not forced to share passwords, a point of significant vulnerability in terms of security and are able to access their part of the shared account from different devices and locations without having the others to authenticate them.

Also possible here is the allowance of different levels of control for different users according to their role within the environment. Nominating certain profiles as administrators or controllers and allowing them more points of control over others who might have fewer responsibilities could be very useful especially when supporting vulnerable populations.

6 CONCLUSION

Most accounts and profiles design currently adopt the single user to single account mapping whereas the reality lays elsewhere. Accounts are often shared by friends, family, housemates or even acquaintances, sometimes individuals resort to having multiple accounts on the same platform, sharing one or more of them with different social groups and sometimes users support others like teenagers and elders on said platforms. Digital identities and user data are spread across these boundaries but technology still provides minimal support to users in the management of these shares.

This research takes up the aforementioned challenge by looking into the practicalities of everyday account sharing. For this, it includes user cohorts and populations of various types associated with sharing such as co-located shares, remotely located shares, sharing within families and users beyond the generic adult cohort, to include teenagers, a population who are active yet often dependent users of accounts.

Our focus groups which investigated user feedback on sharing practices, user agency around sharing and a near-future sharing scenario revealed a set of user challenges around the experience, social implications of sharing and privacy and security concerns, which are current barriers to sharing; and a set of future expectations around experience enhancement, social requirements and user privacy, which are expected to make the practice of sharing more user-friendly.

This research, while in the formative stages currently, intends to build upon these findings by including the ‘maker’ perspective. This would involve a set of expert interviews to understand the practicalities of working with and implementing sharing support on technologies while exploring the landscape in more detail and rigour. We believe this further work would provide the opportunity to delve further into additional areas like the legal implications of sharing (e.g. sharing Netflix passwords has been contentious in the past with claims of it violating the Computer Fraud and Abuse Act (CFAA) in the United States) along with further investigation of phenomena uncovered by this work like social nuance, e.g., designing for hierarchies within accounts and inherent power dynamics. Hence, here, ‘makers’ would include researchers and practitioners with an interest in the domain but working on different dimensions of it like design, technology, data management and legal aspects, to ensure a holistic and comprehensive account of the inherent priorities that require sensitivity and response. We envision that bringing together the views of the user challenges and maker challenges would help produce a set of design guidelines that could form a comprehensive and practical guide towards the integration of support and agency around account sharing.

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